

TorqBall — DIY Build Guide v1

0. What we build, what we buy

Everything the [TorqBall — Venue Scenarios & Build Plan v1](#) §4 table marks **DIY**, turned into a shopping list, a cut list and a step-by-step. Prices are Sept-2026 India ballparks — check the live listing before ordering. Amazon links are **search links**, not fixed listings, so they keep working when a seller changes.

Item	Scenario	Build time (2 people)	Est ₹	Section
Ramp kit — 2 tabletops, 2 rollers, 2 chicane sets	All	1.5 days	5,700–9,700	§2
Goals + LED — 2	All	Half a day	3,800–6,800	§3
Barricade panels — ~28 + 4 corners	A only	3 days, in parallel	22,000–39,000	§4
Soft noodle line — one open side	B, C	1 hour	2,100–3,400	§5
Driver stand + backdrop frame	All	1 day	6,600–12,200	§6
Pit + charging table	All	Buy + 1 hour setup	12,900–24,300	§7
Scoreboard / AV	All	1 hour	0–2,000	§8
Tools, if we own none	—	—	6,000–10,000	§1

Not in this guide (vendor): branding print, DJ/sound, food, inflatable air-wall, tent, portable AC — see the Venue Scenarios doc §4.

1. Tools + consumables — one shopping run

Buy once, reuse every event. Skip anything already in the garage.

Tools

Tool	Why	Est ₹	Where
Cordless drill/driver, 12–18 V, with bit set	Every screw in this guide	2,500–4,500	amazon.in
Jigsaw	Ramp slopes, goal-panel openings, corner curves	2,500–4,500	amazon.in
Hand saw (PVC + battens)	Cuts pipe and pine cleanly without a power tool	300–500	amazon.in

Tool	Why	Est ₹	Where
Measuring tape 5 m + carpenter's square	Layout	300–600	amazon.in
F-clamps, 4 × 300 mm	Hold skins while screwing	800–1,400	amazon.in
Utility knife + spare blades	EVA foam cuts	150–300	amazon.in
Sandpaper 80 + 120 grit	Every cut edge gets sanded — splinters snag cars	200–400	amazon.in
Hot-glue gun + sticks	Tacking LED strip and net	300–600	amazon.in
Safety glasses + dust masks	Ply dust, EVA glue fumes	300–500	amazon.in

Consumables (shared across every build below)

Item	Qty	Est ₹	Where
Wood screws 25 mm + 40 mm, boxes	1 box each	400–700	amazon.in
Fevicol SR 998 contact adhesive (EVA to ply)	1 L	400–600	amazon.in
Fevicol MR / wood glue	500 g	200–300	amazon.in
Cable ties 200 mm + 300 mm, black	100 each	300–500	amazon.in
Gaffer tape, black, 2 rolls	2	500–900	amazon.in
Chevron hazard tape, orange/white	3 rolls	450–900	amazon.in
Matte black spray paint	4 cans	800–1,400	amazon.in
Heavy-duty adhesive velcro, 25 mm	5 m	400–800	amazon.in
Empty sandbags (fill locally)	40	1,500–2,800	amazon.in
Sand	2 bags	200–400	Local hardware / construction supplier

Local, not Amazon: plywood (timber yard, take §9's cut plan), 48 × 24 mm pine battens, 25 mm PVC pipe in 3 m lengths, euro pallets (OLX / industrial packaging suppliers). Every Mumbai timber yard cuts to size for ₹20–40 a cut — let them do the long straight cuts.

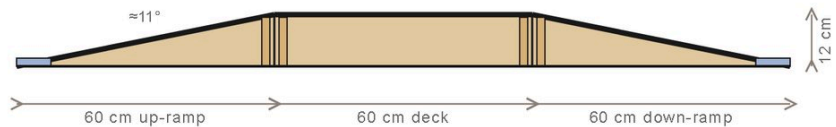
2. Ramp kit

Ramp kit — tabletop, roller, chicane

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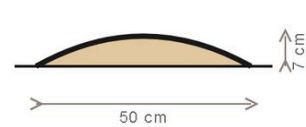
Plywood + pine battens + EVA face + chevron tape · sized for 1/16–1/10 buggies · scattered on the pitch, never a fixed lane

A · TABLETOP — side section (3 modules, hinged or velcro-butted)



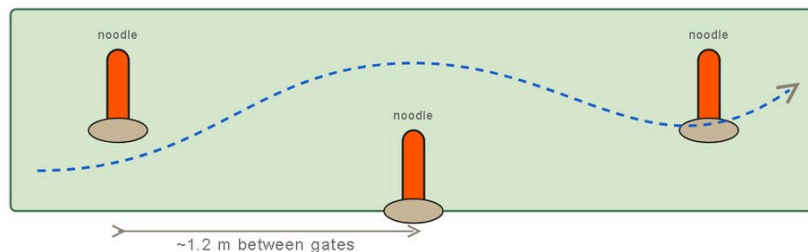
Width 60 cm (two cars side by side won't fit — that's intended)
Top skin 9 mm ply · frame 48×24 mm pine · EVA 10 mm on both lips
Chevron tape on both approach edges so drivers read the lip on camera

B · ROLLER — small hump



4 mm bendy ply over 3 pine ribs
Or two shallow ply slopes meeting at a rounded EVA cap — same footprint

C · CHICANE — top view, pool-noodle gates on sandbag feet



1.2 m pool noodle on a 25 mm PVC stub set in a sandbag / filled bottle
Cars brush it, it gives — no damage
3 gates = one chicane; make 2 sets

9 mm plywood skin

48×24 mm pine batten

10 mm EVA foam

chevron tape / noodle

The ramps are **scattered on the pitch during play**, not a lane and not a zone — the design goal is small, light, soft-edged modules a marshal can pick up and move between matches. Sized for 1/16–1/10 buggies (~1 kg cars, speed-capped per [Car Fleet, Scale & Protection](#)).

What one kit is: 2 tabletops (each = up-ramp + deck + down-ramp), 2 rollers, 2 chicane sets (3 noodle gates each). One kit is plenty for a 6.5 × 10 m pitch; anything more clutters the game.

Materials

Item	Qty	Est ₹	Where
9 mm plywood, 8 × 4 ft	1 sheet (see §9 cut plan)	1,200–1,800	Timber yard
4 mm bendy/flexi plywood, 8 × 4 ft	1 sheet (rollers)	700–1,100	Timber yard
48 × 24 mm pine batten	12 m	600–1,000	Timber yard
EVA foam sheet, 10 mm	1 m ²	500–1,000	amazon.in
Piano hinge, 600 mm	4 (or velcro instead)	400–800	amazon.in
Pool noodles, 1.2 m	6	900–1,500	amazon.in
25 mm PVC pipe, 3 m	1 length + 6 end caps	300–500	Local hardware
Sandbags (from §1 stock)	6	—	—

Item	Qty	Est ₹	Where
Chevron tape, screws, glue (from §1)	—	—	—
Ramp kit total		≈ 5,700–9,700	

Cut list (per tabletop — make 2)

- 9 mm ply: up-ramp skin 600 × 610 mm, deck skin 600 × 600 mm, down-ramp skin 600 × 610 mm (the extra 10 mm on the slopes is the overlap onto the deck edge)
- Pine batten: 2 × 600 mm (deck long sides), 4 × 120 mm (deck legs), 2 × 600 mm (up-ramp sides, cut on the diagonal from 120 mm to 0), same again for the down-ramp
- EVA: 2 strips 600 × 80 mm (the two approach lips)

Steps — tabletop

1. Build the **deck frame** first: two 600 mm battens + four 120 mm legs into a 600 × 600 × 120 mm open box. Glue + two 40 mm screws per joint. Check it sits flat on the floor — rock it, shim any leg.
2. Screw the **deck skin** on with 25 mm screws every 150 mm. Sand every edge to a round-over — this is the edge a car's bumper hits.
3. Cut the **slope sides**: on a 600 mm batten, mark 120 mm at one end and 0 at the other, cut the diagonal. Make 4 (2 per slope). Screw a slope skin onto two slope sides so the tall end is 120 mm and the low end sits flat on the floor. Repeat for the second slope.
4. **Join the slopes to the deck** with a piano hinge on the top face (the kit folds flat for the car) — or, simpler, glue heavy-duty velcro on the deck's vertical edge and the slope's tall end so they butt together and can be pulled apart.
5. Glue a **60 × 8 cm EVA strip** to each approach lip with Fevicol SR — the lip is where a car nose-dives on a bad approach.
6. **Chevron tape** across both approach edges and along the deck's side edges. Drivers read the lip on the overhead camera; so does the crowd.
7. Spray the sides **matte black** — the lit ball reads against it.
8. Roll a car over it at walking pace. If it bottoms out at the transition, the deck frame is too tall for the car — drop the legs to 100 mm.

Steps — roller

1. Cut 3 pine **ribs** from a 600 mm batten offcut, each a shallow arc 500 mm long × 70 mm high at the centre (draw the arc with a bent batten as a guide, cut with the jigsaw).
2. Screw the ribs to two 500 mm base battens, spaced at 0 / 300 / 600 mm.
3. Wet the **4 mm bendy ply** skin (500 × 600 mm) lightly, bend it over the ribs, clamp, screw down every 100 mm along each rib. Let it dry overnight before unclamping.
4. Sand, EVA strip on both edges, chevron tape, matte black. Same test as the tabletop.

Steps — chicane gate (make 6)

1. Cut 25 mm PVC into 6 × 400 mm stubs, cap one end.
2. Fill a sandbag to ~2 kg, push the capped stub into the middle of it so it stands upright, tie the bag off around it.

3. Slide a pool noodle over the stub. Done — the noodle folds when a car hits it and springs back.
4. Three gates 1.2 m apart in a zig-zag = one chicane. Move them every few matches so the pitch stays fresh.

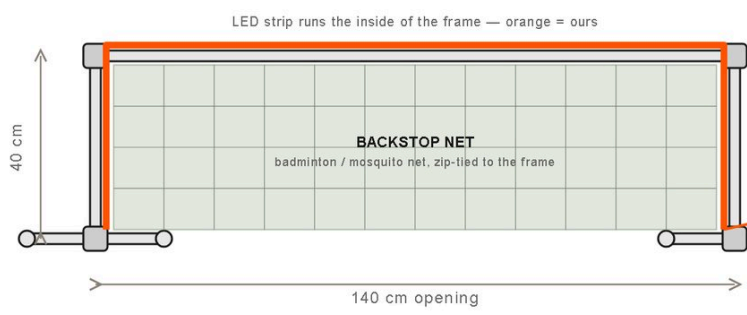
3. Goals + LED

Goal — PVC frame, LED strip, backstop net

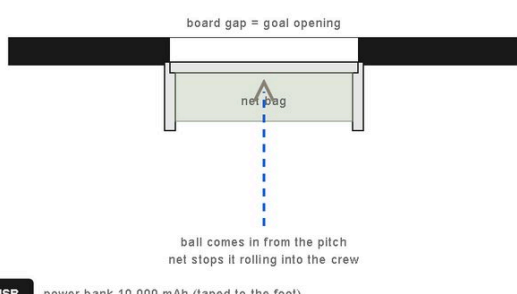
2 goals · 1.4 m opening × 40 cm tall (Rulebook §3) · 25 mm PVC, USB 5 V LED strip on a power bank, no soldering



A · FRONT VIEW



B · PLAN VIEW — sits in the board / air-wall gap



PVC CUT LIST — per goal (25 mm pipe, 2 × 3 m lengths)

Uprights	2 × 40 cm
Crossbar	1 × 140 cm
Feet	2 × 30 cm
Elbows 90°	2
Tees	2 (foot to upright)
End caps	4

Decathlon alternative: a 5-a-side pop-up goal pair (~₹2–4k) — zip-tie the same LED strip to it. Compare before cutting pipe. Feet get a sandbag each so a car nudge doesn't walk the goal.

Two goals, **1.4 m opening × 40 cm tall** ([TorqBall — Rulebook §3](#)). A PVC frame with a USB LED strip on a power bank — no soldering, no mains near the pitch. A backstop net stops a scored ball rolling into the crew.

Decathlon alternative: a 5-a-side pop-up goal pair (~₹2,000–4,000) with the same LED strip zip-tied on. Faster, slightly less on-brand. Price both before cutting pipe — if the pop-up pair is under ₹3k, buy it.

Materials (for 2 goals)

Item	Qty	Est ₹	Where
25 mm PVC pipe, 3 m	4 lengths	500–800	Local hardware
25 mm PVC 90° elbows	4	100–200	amazon.in
25 mm PVC tees	4	100–200	amazon.in
25 mm PVC end caps	8	100–150	amazon.in

Item	Qty	Est ₹	Where
USB 5 V LED strip, 5 m, single colour (orange/white)	2	600–1,200	amazon.in
Power bank, 10,000 mAh	2 (or reuse ours)	1,600–3,000	amazon.in
Badminton net or fine garden net	1 (cut in half)	300–600	amazon.in
Sandbags (from ₹1 stock)	4	—	—
Cable ties, gaffer tape (from ₹1)	—	—	—
Goals total		≈ 3,800–6,800	

Optional upgrade for content: an addressable WS2812B strip + ESP32 so the goal flashes on a score — the build for that is in [TorqBall — Ball & Boards Test](#) §Boards BOM. Not needed for event #1.

Cut list (per goal)

- 2 × 400 mm uprights · 1 × 1,400 mm crossbar · 2 × 300 mm feet (each foot cut into 2 × 150 mm halves, joined by a tee)

Steps

1. **Dry-fit** the frame: foot tee → upright → elbow → crossbar → elbow → upright → foot tee. Feet run front-to-back so the goal can't tip forward when the ball hits the net. Don't glue — friction-fit PVC comes apart for transport.
2. Stand it in the goal gap of the boards (Scenario A) or on the goal line (B, C). Check the 1.4 m opening with a tape.
3. **LED strip:** peel-and-stick along the *inside* face of the two uprights and the crossbar (the face that looks at the pitch). Hot-glue every 30 cm — the adhesive alone lets go on PVC. Route the USB lead down one upright to the foot.
4. Gaffer-tape the **power bank** to the outside of that foot. 5 m of 5 V strip draws ~1–2 A; a 10,000 mAh bank runs it all day.
5. **Net:** cut the badminton net to ~1.5 × 1.0 m, zip-tie it to the crossbar and both uprights, let it hang loose behind to make a bag. It only needs to stop a light foam ball.
6. One **sandbag on each foot**. A car nudging the post must not walk the goal.
7. Test: roll the ball in at full speed. If it bounces back out of the net onto the pitch, the net is too tight — slacken it.

4. Barricade panels — Scenario A only

Barricade panel — Scenario A only

1.2 m × 40 cm modules · 6–9 mm ply + 10 mm EVA face · foot bracket, carriage bolt + wingnut joints · ~28 panels for the full perimeter

A · EXPLODED — one panel

120 cm

40 cm

6–9 mm PLYWOOD 120 × 40 cm · 4 × M8 holes

10 mm EVA FOAM face — glued (Fevicol SR) or heavy-duty velcro strips
this is the side the cars and the ball hit

FOOT — 60 × 12 cm ply plate screwed to a 48×24 mm pine batten on the back
+ 1 sandbag (2–3 kg) on the plate so a car hit doesn't walk it

B · JOINT — panel to panel (tool-free)

M8 × 50 carriage bolt through both panels, wingnut on the back
2 per joint · seen from the back · EVA faces meet on the pitch side

C · CORNER — never square

YES
4 mm bendy ply on a 30 cm radius,
wrapped in 2 layers of EVA — or butt two
straights + a pool noodle zip-tied in the angle

NO
square corner — the ball jams,
a chassis snags, a hand catches

6–9 mm plywood

10 mm EVA foam

pine batten / foot

wingnut

The one real carpentry job, and only if the venue is a raw hall with nothing to contain the pitch. Spec is the locked one from [TorqBall — Ball & Boards Test](#) §Boards: **1.2 m × 40 cm panels, 6–9 mm ply, 10 mm EVA face, radius corners, no floor drilling.** ~33 linear metres of perimeter minus two goal gaps = **~28 straight panels + 4 corners.**

Do the Ball & Boards Test first — build 4–6 panels, run the rebound and car-hit checks in that doc, then cut the rest. Don't cut 30 panels on day one.

Materials

Item	Qty	Est ₹	Where
6–9 mm plywood, 8 × 4 ft	7 sheets (5 panels + 1 goal/corner + 1 feet — §9)	9,800–15,400	Timber yard
4 mm bendy plywood, 8 × 4 ft	1 sheet (4 radius corners)	700–1,100	Timber yard
EVA foam sheet, 10 mm	~13 m ²	6,500–13,000	amazon.in
M8 × 50 carriage bolts	60	900–1,800	amazon.in
M8 wingnuts + washers	60 each	600–1,200	amazon.in

Item	Qty	Est ₹	Where
48 × 24 mm pine batten (feet)	18 m	900–1,500	Timber yard
Sandbags (from §1 stock, 2–3 kg each)	30	—	—
Matte black paint (roller + tray, not spray at this scale)	2 L	900–1,600	amazon.in
Fevicol SR, screws, sandpaper (from §1, buy double)	—	1,500–2,500	—
Barricade total		≈ 22,000–39,000	

Lower than the ₹30–54k in the Ball & Boards BOM because 6 panels come out of one sheet on the §9 cut plan — the earlier estimate assumed 8–10 sheets.

Steps — straight panel (make ~28)

1. Timber yard cuts every sheet into **6 × 1,200 × 400 mm** (cut plan §9). Bring the plan.
2. Drill **4 × 9 mm holes** per panel: 40 mm in from each side edge, at 80 mm and 320 mm from the bottom. Use one drilled panel as the template for all the rest so every joint lines up.
3. Sand every edge. Splinters here snag a chassis and cut a marshal's hand.
4. **Foot:** screw a 600 mm pine batten flat along the back face, 20 mm up from the bottom edge. Screw a 600 × 120 mm ply plate (from sheet 7) to the batten so it sticks out backwards like an L. One sandbag sits on the plate on event day.
5. **Paint** the back face and edges matte black. Two coats.
6. **EVA face:** cut EVA to 1,200 × 400 mm, coat both surfaces with Fevicol SR, wait 10 minutes till touch-dry, press together from one edge. Roll flat with a bottle. Masks on — the fumes are real.
7. Trim the EVA flush. The finished panel is 1,200 × 400 × ~20 mm and weighs ~4 kg.

Steps — corner (make 4)

1. Cut 4 mm bendy ply to **1,000 × 400 mm**. Screw it to two short battens at each end so it holds a **30 cm radius** curve (soak the inside face lightly if it fights you).
2. Two layers of EVA on the pitch face — corners take the hardest hits.
3. Drill the same 4-hole pattern at each end so it bolts to a straight panel.
4. **Fallback if bendy ply is a fight:** butt two straight panels at 90° and zip-tie a full pool noodle vertically into the inside angle. Uglier, works.

Steps — goal panels (4)

1. From sheet 6, cut 4 panels **900 × 400 mm** instead of 1,200. Two per short end, one each side of the goal, leaving the **1,400 mm gap** for the §3 goal frame.
2. Same holes, foot, paint, EVA. Wrap the cut edge facing the goal in EVA too.

Assembly on the day (2 people, ~45 min)

1. Lay a tape line for the 6.5 × 10 m pitch. Set panels foot-outward along it, EVA face in.

2. Bolt each joint: 2 carriage bolts through from the pitch side, washer + wingnut on the back. Hand-tight — the bolts locate, the sandbags hold.
3. Corners last. Drop a sandbag on every foot. Walk the perimeter kicking each panel — nothing should shift.

5. Soft noodle line — Scenarios B and C

On a turf, the walls or the inflatable air-wall contain three sides; the driver-stand side just needs a **visual line the cars respect**. Pool noodles on the floor do it.

Item	Qty	Est ₹	Where
Pool noodles, 1.2 m	10	1,500–2,500	amazon.in
25 mm PVC pipe, 3 m	1 length	150–250	Local hardware
Sandbags or 1 L filled bottles	10	300–600	—
Noodle line total		≈ 2,100–3,400	

1. Cut PVC into 10 × 300 mm stubs. Slide each noodle onto a stub so the stub sits inside one end.
2. Lay noodles end to end along the driver-stand side of the pitch, stub end pushed into a sandbag or under a filled bottle every 1.2 m.
3. Gaffer-tape the joints. Cars bounce off; the ball rolls over — that's fine, it's a driver line, not a wall.

6. Driver stand + backdrop frame

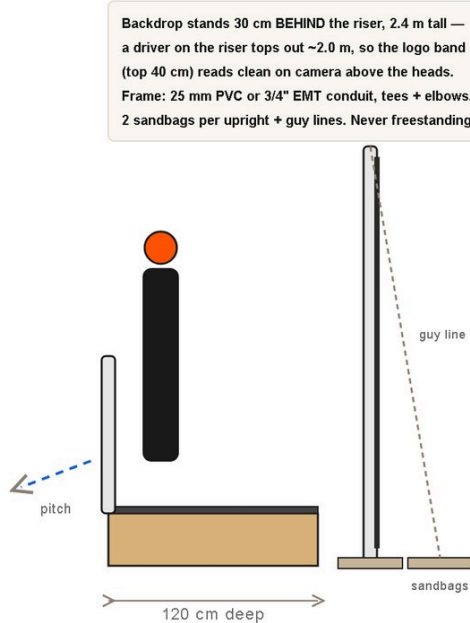
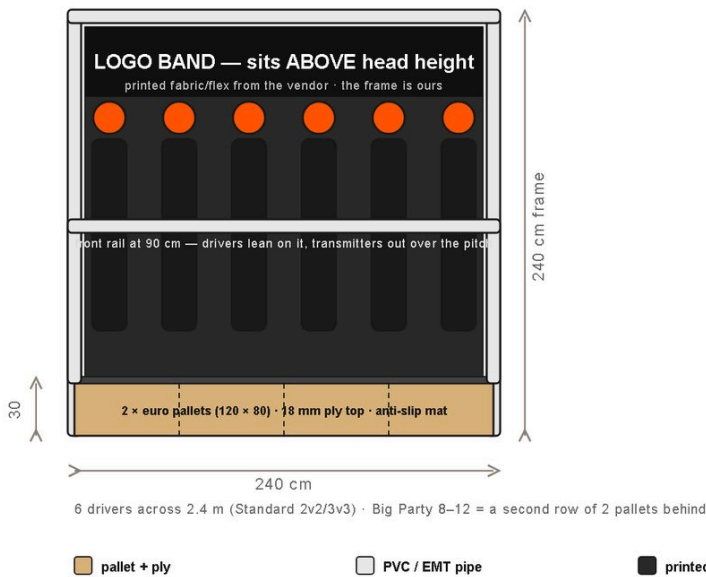
Driver stand + backdrop frame

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Pallet riser 2.4 × 1.2 m, 30 cm high · PVC/EMT backdrop frame 2.4 × 2.4 m · print (vendor) zip-ties to the frame

A · FRONT VIEW (from the pitch)

B · SIDE VIEW



A **2.4 × 1.2 m riser, 30 cm high**, sized to 6 drivers across (Standard 2v2/3v3). Big Party's 8–12 gets a second row of two pallets behind. The **backdrop frame is ours; the print is the vendor's** — send them the frame size (2.4 × 2.4 m) and ask for hemmed edges with eyelets every 30 cm.

Materials

Item	Qty	Est ₹	Where
Euro pallets, 1,200 × 800 mm, good condition	2 (4 for Big Party row)	1,000–3,600	OLX / packaging suppliers, local
18 mm plywood, 8 × 4 ft	1 sheet, uncut	2,200–3,000	Timber yard
Rubber anti-slip mat, 2.4 × 1.2 m	1	1,200–2,000	amazon.in
L-brackets 50 mm + screws	8	300–500	amazon.in
25 mm PVC pipe, 3 m (frame + rail)	7 lengths	1,000–1,700	Local hardware
25 mm PVC tees + elbows	8 + 6	300–500	amazon.in
Sandbags (from \$1 stock)	6	—	—
Nylon rope / guy line, 20 m	1	200–350	amazon.in

Item	Qty	Est ₹	Where
Stand total		≈ 6,600– 12,200	

Stage-deck rental fallback: ~₹800–1,600/day if the pallets don't turn up in time. Same footprint.

Steps — riser

1. Set the 2 pallets side by side on the floor, long sides touching, so the footprint is 2,400 × 800 mm. Check the top is level — shim a corner with a ply offcut if it rocks.
2. Screw the 2 pallets together through the side stringers with 4 × L-brackets.
3. Lay the **18 mm ply sheet** on top (2,440 × 1,220 — it overhangs the 800 mm pallet depth by 400 mm; that overhang is toward the *back*, supported by a 48 × 24 batten leg at each end). Screw down through the ply into every pallet top-board, 40 mm screws.
4. Sand the front edge round. Chevron-tape it — it's a trip edge in a dark room.
5. Lay the **anti-slip mat**, gaffer-tape the edges down.
6. Jump on it. If anything flexes, add a batten leg under the middle of the overhang.

Steps — front rail

1. PVC: 2 × 900 mm uprights, 1 × 2,400 mm rail (join two lengths with a straight coupler), 2 elbows.
2. Fix the uprights to the front corners of the riser with a tee screwed into the ply. Drivers lean on this with their transmitters over the pitch — friction-fit is not enough, screw it.

Steps — backdrop frame

1. PVC: 2 × 2,400 mm uprights (a 3 m length cut down, or EMT conduit which is stiffer), 1 × 2,400 mm top bar, 2 elbows, 2 tees at the bottom into 2 × 600 mm feet each.
2. Stand it **30 cm behind the riser**. Two sandbags on each foot.
3. **Guy lines** from each top corner back to a sandbag on the floor — the frame is a sail in a hall with AC blowing and a crowd brushing past. Never leave it freestanding.
4. Zip-tie the printed backdrop through its eyelets to the top bar and uprights. Pull it taut top-first so it hangs flat on camera.
5. Check the shot from the overhead/hero camera position: logo band above the drivers' heads, no upright visible through a gap in the print.

7. Pit + charging table

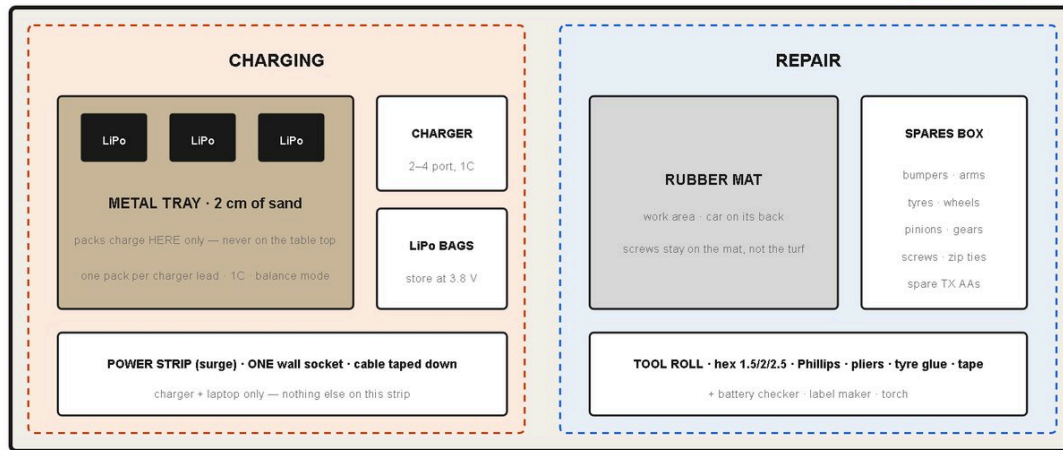
Pit + charging table — layout

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One 6 ft folding table · charging on the LEFT, repair on the RIGHT · nothing charges on turf, ever

6 ft × 2.5 ft folding table (180 × 75 cm) — plan view

pit-cam (optional)



ON THE FLOOR, at the charging end



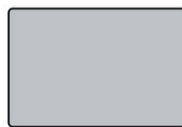
ABC extinguisher
2 kg



metal bin + lid
(puffed pack goes in)



bucket of sand



1 m² cement board / metal
sheet under the tray end

FLOOR under the charging end must be bare tile, concrete, or the cement board — never turf or ply.
One person = LiPo warden. Never unattended.
Charge log on a clipboard: pack # · in · out · V.

Mostly a **buy list**, not a build. The one rule that isn't negotiable: **LiPo packs charge on a metal tray of sand, on a bare floor, watched by one named person.** The rest of the protocol is in [Event Ops Standards \(all games\)](#). Charger, spares and pack count depend on the rental fleet — confirm with the fleet supplier what comes with the cars before buying.

Item	Qty	Est ₹	Where
Folding table, 6 ft	1 (or borrow)	3,000–5,000	amazon.in
LiPo-safe charging bags	4	800–1,600	amazon.in
Balance charger, 2–4 port (if the fleet doesn't bring one)	1	2,500–5,000	amazon.in
Steel tray, ~50 × 35 cm	1	400–800	amazon.in
ABC dry-powder extinguisher, 2 kg	1	1,200–2,000	amazon.in
Metal bin with lid	1	500–900	amazon.in
Surge-protected power strip, 6 socket, 3 m lead	1	800–1,500	amazon.in
Rubber work mat, 60 × 40 cm	1	400–800	amazon.in

Item	Qty	Est ₹	Where
RC hex driver set 1.5 / 2.0 / 2.5 + Phillips	1	500–1,200	amazon.in
LiPo voltage checker / cell meter	1	300–600	amazon.in
Cement board or steel sheet, ~1 m ² (floor under the tray end)	1	400–800	Local hardware
Sand (from \$1 stock), clipboard, marker	—	100–200	—
Spares box — bumpers, arms, tyres, pinions, screws (fleet-specific)	1	2,000–4,000	Fleet supplier / amazon.in
Pit total		≈ 12,900–24,300	

Setup (1 hour, on the day)

1. Table goes on **bare floor** — pull the turf back or lay the cement board under the charging end. Never on turf, never on the ply riser.
2. Charging end: tray with 2 cm of sand → packs lie on the sand while charging → charger beside it → LiPo bags for anything not charging. Extinguisher, sand bucket and metal bin on the floor at that end.
3. One power strip, one wall socket, cable gaffer-taped to the floor. Charger + laptop only on that strip.
4. Repair end: rubber mat, spares box, tool roll. Screws stay on the mat.
5. Name the **LiPo warden** at the crew brief. They log every pack in/out with voltage. A puffed, hot or hissing pack goes straight into the metal bin with the lid on and outside.

8. Scoreboard / AV

Cheapest possible: a laptop and a TV we already own.

Item	Qty	Est ₹	Where
HDMI cable, 5 m	1	300–500	amazon.in
TV (32–43", owned or borrowed) + its stand	1	0	—
Bluetooth speaker or the DJ's PA for the hooter	—	0	DJ vendor
Scoreboard: any free browser scoreboard, or a Google Slide per match	—	0	—
AV total		≈ 0–2,000	

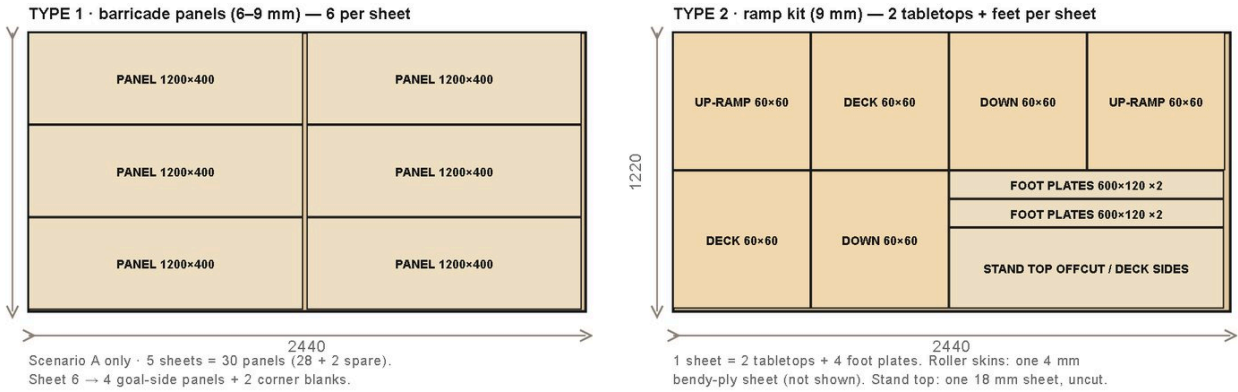
1. TV on its stand beside the MC position, facing the crowd, above head height (Scenario A: the position marked on the floor plan above the pitch).
2. Laptop on the pit table's dry end or the MC stand, HDMI to the TV. The MC or a marshal ticks the score — it does not need to be automated for event #1.
3. Hooter = a 2-second sound file the MC plays through the PA at full time. Test it with the DJ at sound check.

9. Plywood cut plan

Plywood cut plan — 8 × 4 ft sheets

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2440 × 1220 mm sheets · take this to the timber yard, most Mumbai ply shops cut for ₹20–40 a cut · leave 3 mm kerf between pieces



Take this to the timber yard on D5 morning. Buy 8 × 4 ft sheets (2,440 × 1,220 mm) and have them cut the long straight cuts on the panel saw — cleaner than a jigsaw and ₹20–40 a cut.

Sheet	Thickness	Yields	Scenario
Type 1 × 5	6–9 mm	30 barricade panels, 1,200 × 400 (28 + 2 spare)	A only
Type 1 × 1 more	6–9 mm	4 goal-side panels 900 × 400 + 2 corner-end blanks	A only
Type 1 × 1 more	6–9 mm	40 foot plates, 600 × 120 (28 needed)	A only
Type 2 × 1	9 mm	2 tabletops (6 × 600 mm skins) + 4 ramp foot plates	All
Bendy × 1	4 mm	2 roller skins + 4 radius corners (A)	All / A
Stand top × 1	18 mm	uncut, 2,440 × 1,220	All

Totals: Scenarios B/C = 3 sheets (9 mm, 4 mm bendy, 18 mm). Scenario A = 10 sheets.

10. Build order — D5 to D8, 2–3 people

Slots into the [TorqBall — Venue Scenarios & Build Plan v1](#) §5 timeline. Amazon orders go in on **D4 evening (18 Sep)**, the moment the venue is locked, so everything lands by D6.

Day	Date	Morning	Afternoon	Needs
D5	19 Sep	Timber yard with the §9 cut plan · hardware shop for PVC, battens, sand	Ramp deck frames + slope sides cut and glued · goal PVC dry-fit	Drill, jigsaw, clamps

Day	Date	Morning	Afternoon	Needs
D6	20 Sep	Skin the tabletops, EVA lips, chevron · roller ribs + bendy ply (clamp overnight)	Goals: LED strip, net, power banks · chicane gates · Amazon delivery check	Hot glue, Fevicol SR
D7	21 Sep	Riser: pallets, ply top, mat, rail · backdrop frame + guy-line test	Noodle line kit (B/C) · pit kit unboxed and labelled · paint everything matte black	Both people on the riser for the ply lift
D8	22 Sep	Dry-fit in the garage: goals + ramps + stand + pit table as one layout · fix what's wrong	Pack everything into labelled crates · charge the power banks · print vendor confirms D10 delivery	—

Scenario A adds a third person D5–D8 full-time on panels: D5 drill + sand all 34 · D6 feet + paint · D7 EVA faces (fumes — do it outdoors or with the garage open) · D8 corners + goal panels + a test bolt-up of 6 panels.

D10 (24 Sep) is the full rehearsal — everything built here goes up once, at full size, before the venue.

11. Cost roll-up

	Scenario A — raw hall	Scenario B — AC turf	Scenario C — outdoor turf
Ramp kit (§2)	5,700–9,700	5,700–9,700	5,700–9,700
Goals + LED (§3)	3,800–6,800	3,800–6,800	3,800–6,800
Barricade panels (§4)	22,000–39,000	—	—
Noodle line (§5)	—	2,100–3,400	2,100–3,400
Driver stand + frame (§6)	6,600–12,200	6,600–12,200	6,600–12,200
Pit + charging (§7)	12,900–24,300	12,900–24,300	12,900–24,300
Scoreboard / AV (§8)	0–2,000	0–2,000	0–2,000
DIY build total	≈ 51,000–94,000	≈ 31,000–58,000	≈ 31,000–58,000
Tools if none owned (§1)	+6,000–10,000	+6,000–10,000	+6,000–10,000

Everything here is a **one-time build** that travels to the next event. The pit list drops by ₹5–9k if the fleet rental brings its own charger and spares. Vendor lines (print, DJ, food, air-wall, tent, AC) sit on top — see the Venue Scenarios doc §4.

12. Safety while building

- Glasses and mask for every cut and every glue-up. Fevicol SR fumes need open air.
- Sand every ply edge before it goes near a car, a ball or a hand — the boards rule in the Rulebook (§10, no exposed edge or fastener) starts in the garage.
- Carriage bolts go in from the pitch side so the smooth dome faces the cars; wingnuts stay on the crew side.

- Nothing electrical is DIY here beyond a USB LED strip on a power bank. Mains stays with the venue's socket and one surge strip.
- LiPo protocol is not a build step — it's a person. Name the warden before anything charges.